

Robust Testing in Automotive Safety: Designing Critical Edge Case Scenarios

Kukadiya Romikumar Rameshbhai
Faculty of Mechanical and Mechatronics
Deggendorf Institute of Technology
Deggendorf, Germany
romikumar.kukadiya@stud.th-deg.de

Abstract—The rapid development of automated vehicles (AV) promises to make transportation both safer and more efficient. However, their reliability in the real world in diverse conditions is a big challenge, especially due to rare and unexpected situations known as edge cases. There is no universal validation and/or verification mechanism to test AV in these cases currently. The standards rely on the approach where testing is done through defined sets of cases to address specific requirements. This paper focuses on methods to test AVs in critical edge scenarios involving snowy conditions. These conditions pose a risk for automotive safety systems. By using YOLO object detection algorithm to test the ability of AV in detection of pedestrian under these adverse conditions. The findings will help to highlight vulnerabilities in the current testing process and contribute to the development of robust testing methodologies.

Keywords— *Automated Driving, Automotive Safety, Critical Edge Scenarios, Object Detection, YOLO Algorithm*

I. INTRODUCTION

Autonomous vehicles (AVs) have seen rapid advancements in last few years, but their widespread adoption remains limited by many challenges, especially in the areas of testing and validation. The testing process for AV systems is quite complex, having not only technical trials in testing environments but also on-road testing in a wide range of road and weather conditions including unexpected and critical scenarios. There are numerous scenarios that AVs may come across, but it is very difficult to predict and test the AVs in all these conditions. This limitation shows the role of Edge Cases in the development of AV.

Edge cases are important during the development and testing of autonomous vehicles (AVs) for several important reasons. Edge case scenarios are normally the difference between common operation and unexpected scenarios that leads to failure or unsafe behavior in vehicle driving. Testing edge cases helps the AVs prepare better for the diversity of real-world driving. Additionally, regulatory bodies and standardization organizations are establishing safety standards for autonomous vehicles (AVs), requiring them to prove their ability to manage a wide range of scenarios. However, these standards often have limitations, as they may not fully test the complexity of all potential edge cases or the nature of real-world driving conditions.

The levels of driving automation in SAE J3016 defines the SAE Levels from Level 0 (no driving automation) to Level 5 (full driving automation) in the context of Autonomous vehicles and their operation on roadways, with levels 3–5 being classified under the Automated Driving Systems (ADS) category [1]. Autonomous Vehicles with higher levels of automation, generally SAE level 3–4 equipped with ADS, have been tested in various parts of the world including Germany, U.S., and China, to evaluate their safety and reliability.

Automated driving testing is primarily conducted through two core methods: real-world testing and simulation testing.

A. Real World Testing

The real world testing is done by majority of developers. And there are two types of real world testing mentioned: Scenario based testing and On road testing [2]. In scenario based testing specific driving scenarios in closed automated vehicle proving grounds. It includes common driving scenarios such as changing lanes, following other cars, and turning left or right, as well as special road sections such as intersections and roundabouts. Finally, dangerous driving scenarios such as emergency braking and collision are also conducted. In on-road testing, autonomous vehicles are tested in public roads where various scenarios and unexpected conditions could occur. Waymo conducted over 1 million kilometres of on-road testing in 2020 [3].

B. Simulation Testing

Modern simulation environments are powerful enough to test corner cases that are costly to set up in the real world. Safety considerations make simulation testing much more desirable with respect to dangerous driving scenarios, such as collisions. By using the simulation platform, industrial developers do not need to deploy each case of ADS on the real vehicle and construct real-world test scenarios.

A fundamental requirement for safe and reliable autonomous vehicle operation is the critical function of real-time object detection. By accurately identifying and tracking vehicles, pedestrians, traffic signs, and other objects in the surrounding environment, autonomous vehicles can make positive decisions, navigate complex

traffic scenarios, and mitigate potential hazards. Moreover, object detection is essential for advanced driver assistance systems (ADAS), which provide drivers with essential information and support, ultimately contributing to safer roadways. Among various object detection algorithms, You Only Look Once (YOLO) has become very popular in the automotive field due to its focus on real-time performance [4].

This paper examines the importance of designing and testing edge case scenarios, with a focus on conditions such as snowy weather, which poses unique challenges for AVs, including sensor limitations, poor visibility, and difficulty in detecting pedestrians. By using the application of advanced technologies like the YOLO object detection algorithm, this study aims to assess the ability of AVs to handle edge cases and contribute to more comprehensive testing methods that improve the safety and reliability of autonomous driving systems.

II. LIMITED TESTING SCENARIOS

While the Euro NCAP protocols provide a robust framework for evaluating AEB and LSS systems, their limitations suggest a need for improvement in real-world test scenarios. The test protocol focuses on specific and predefined scenarios (e.g., car-to-pedestrian, car-to-bicyclist) that may not fully encompass the wide variety of real-world interactions between vehicles and vulnerable road users. Tests are conducted on smooth, dry, and uniform surfaces with minimal interference from environmental factors (e.g., no precipitation, wind speeds below 10 m/s, or highly reflective surfaces). Extreme weather conditions such as heavy rain, snow, or fog, which significantly affect system performance, are not included. While some tests include night scenarios, they require specific lighting conditions, which may not replicate real-life low-light variability. The test tracks use standardized lane widths and markings, which may not represent varied road geometries or degraded lane markings encountered in real-life conditions. The targets are dressed in predefined, uniform clothing: Black shirt and blue trousers [5]. This standardization does not account for the diversity in real-world clothing colors, patterns, or materials, which could impact sensor detection, especially vision-based systems. This study aims to bridge this gap by examining how clothing variation affects YOLO's detection accuracy.



Figure 1: Euro NCAP VRU Targets [5]

III. METHODOLOGY

This section details the methodology for testing the robustness of YOLO (You Only Look Once) object detection algorithm in automotive safety. The testing is designed to simulate critical edge case scenarios by adding different clothing types, environmental conditions (day/night), and snowy weather conditions and varying distances.

A. Test Setup

1) *YOLO Algorithm Configuration*: YOLOv11 is selected for its high accuracy and real-time processing capabilities. YOLOv11 is particularly suited for scenarios where rapid and precise detection of objects is critical, ensuring efficient handling of varying environmental and object-based challenges.

2) *Target Clothing*: The test focuses on detecting individuals wearing low-contrast jackets and trousers. These outfits are specifically chosen to mimic real-world scenarios on a snowy day, where the clothing blends into the environment. The selected attire includes a combination of grey, black, and white jackets, paired with trousers of similar tones. This setup challenges YOLOv11's ability to detect objects with minimal contrast against the snowy background.

3) *Lighting Conditions*: The testing environment accounts for two distinct lighting scenarios to evaluate YOLOv11's performance under diverse illumination:

- a) *Daytime*: Testing takes place in natural daylight, which includes conditions such as overcast skies and shadows cast by nearby objects. This helps simulate real-world variations in daytime lighting.
- b) *Nighttime*: Low-light conditions are simulated with varying streetlight arrangements. These may include areas with adequate lighting, dimly lit sections, or unevenly distributed light sources.

4) *Test Environment*:

- a) *Location*: The experiments are conducted in an open and controlled area chosen for its accessibility and ability to simulate various real-world conditions in Deggendorf.
- b) *Camera*: The camera used for testing is an iPhone 16, mounted on a sturdy tripod for stability during data collection.

5) *Weather Conditions*: Testing takes place under snowy weather conditions, which introduce additional challenges for object detection. The conditions include both Falling snow, which may obscure visibility and create visual noise and Piled-up snow, which reduces contrast between the target objects (clothing) and the environment.

6) *Varying Distance*: The tests are conducted across a range of distances, from 5 meters to 25 meters.

Figures 2 to 7 illustrate various test scenarios conducted as part of the study. These scenarios encompass different environmental conditions, clothing types, and distances to evaluate the robustness of the YOLO object detection algorithm.



Figure 2: Detection of a person in a black jacket at 5m during daytime

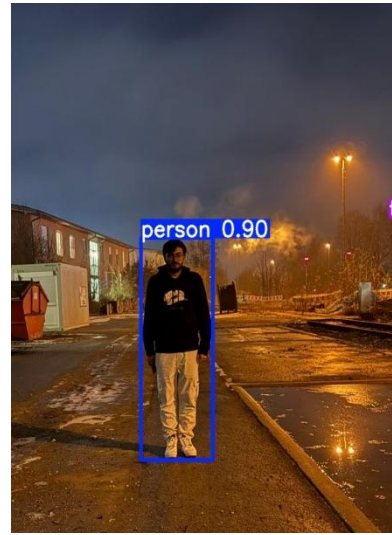


Figure 5 Detection of a person in a black jacket at 5m during nighttime

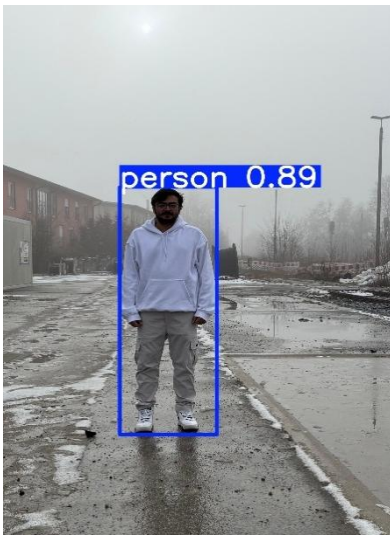


Figure 3: Detection of a person in a white jacket at 5m during daytime



Figure 6 Detection of a person in a white jacket at 5m during nighttime



Figure 4: Detection of a person in a gray jacket at 5m during daytime



Figure 7 Detection of a person in a gray jacket at 5m during nighttime

IV. RESULTS AND DISCUSSIONS

Table 1 provides a comparison of YOLO object detection algorithm's performance in two distinct lighting conditions: daytime and nighttime, with varying object colors (white, black, and gray) and distances (from 5 meters to 25 meters). Below, we will discuss the algorithm's performance in both lighting conditions in detail, breaking down how these factors influence detection accuracy.

A. Daytime Performance:

Fig 8 showcases the relationship between distance and confidence scores of the object detection algorithm in daytime conditions. The confidence scores, which range from 0 (no confidence) to 1 (high confidence), reflect the certainty of the algorithm in successfully detecting the object. The data reveals a general trend of declining confidence scores as the distance increases, highlighting the challenges faced by the algorithm in detecting objects farther from the camera due to reduced object size, lower pixel resolution, and diminished contrast.

Among the three clothing colors, white clothing consistently outperforms black and gray, with confidence scores remaining relatively high and stable across all distances. Starting with a score of 0.89 at 5 meters, the detection confidence for white decreases gradually, reaching 0.68 at 20 meters and recovering slightly to 0.71 at 25 meters. This can be attributed to white clothing's strong light reflectivity and higher contrast against most backgrounds, which enhances its visibility and detectability by the algorithm.

Black clothing, on the other hand, shows strong initial performance with the highest score of 0.91 at 5 meters but experiences a steady decline as the distance increases. By 20 meters, the confidence drops to 0.69, and 0.63 at 25 meters. Black clothing absorbs more light and provides less contrast, it becomes progressively harder for the algorithm to distinguish at greater distances.

Gray clothing emerges as the most challenging color for the algorithm to detect, with a confidence score that starts at 0.89 at 5 meters but deteriorates significantly as the distance increases. By 15 meters, the score drops to 0.75, and by 25 meters, it reaches the lowest value of 0.55 among all tested scenarios. Gray clothing's moderate reflectivity and neutral contrast often blend with various backgrounds, especially in suboptimal lighting conditions, resulting in a dramatic decline in detection performance.

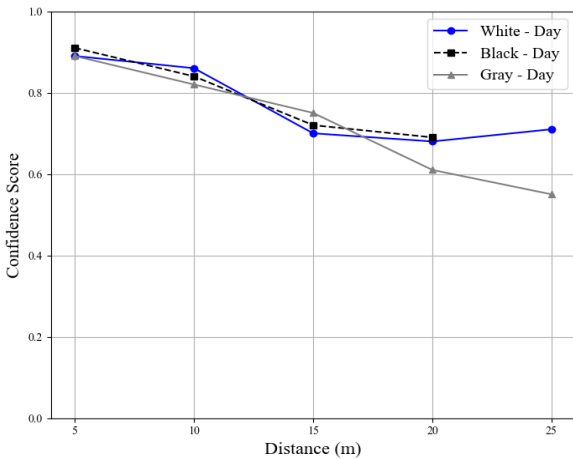


Figure 8: Confidence Score vs Distance Graph for Day

Table 1 YOLO object detection performance

Distance (m)	Confidence Score					
	Day			Night		
	White	Black	Gray	White	Black	Gray
5	0.89	0.91	0.89	0.92	0.90	0.91
10	0.86	0.84	0.82	0.80	0.84	0.80
15	0.70	0.72	0.75	0.69	0.80	0.66
20	0.68	0.69	0.61	0.48	0.67	0.67
25	0.71	0.63	0.55	0.36	0.46	0.42

B. Nighttime Performance: The performance of the object detection algorithm at night varies significantly across different clothing colors—white, black, and gray—highlighting distinct trends and challenges at various distances shown in fig 9..

White clothing while initially performing strongly due to its reflective properties, shows a more pronounced decline in detectability as distance increases. At 5 meters, white clothing achieves the highest score of 0.92, reflecting excellent detectability at close range. However, this advantage diminishes quickly as distance grows. By 10 meters, the score drops to 0.80, and by 15 meters, it falls further to 0.69, indicating a significant loss in detectability compared to black clothing. At 20 meters, white clothing experiences a sharp decline to 0.48, and at 25 meters, it matches gray clothing at a low score of 0.36. The steep drop in confidence suggests that while reflective properties provide an advantage at close ranges, they become less effective at longer distances in low-light conditions.

Black clothing is the most consistently detectable color across various distances, contrary to expectations. At 5 meters, black clothing achieves a high confidence score of 0.90, slightly below white (0.92). Remarkably, as distance increases to 10 meters, black clothing maintains a confidence score of 0.84, outperforming white clothing, which drops to 0.80. This trend continues at 15 meters, where black clothing achieves a score of 0.80, significantly higher than white (0.69) and gray (0.66). Even at 20 meters, black clothing maintains a respectable confidence score of 0.67, again surpassing white

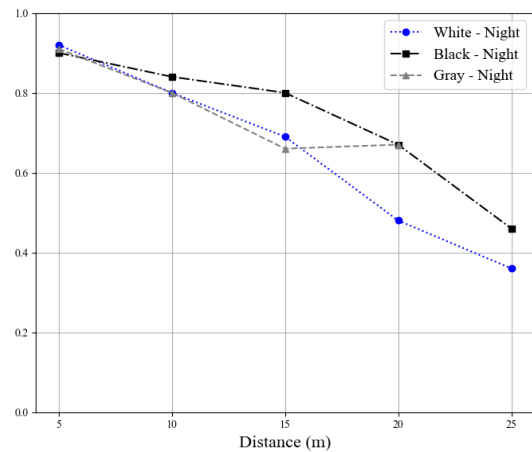


Figure 9: Confidence Score vs Distance for Night

(0.48) and matching gray. At the furthest distance of 25 meters, black clothing retains the highest detectability among all colors with a confidence score of 0.46, while white and gray plummet to 0.36. and 0.42 respectively. These results indicate that, surprisingly, black clothing proves to be the most consistently detectable color under nighttime conditions, likely to benefit from specific contrasts with certain dark or dimly lit backgrounds.

Gray clothing consistently demonstrates the weakest performance among the three colors under nighttime conditions. At 5 meters, gray clothing achieves a high confidence score of 0.91, similar to the other colors, but its detectability rapidly declines as distance increases. By 10 meters, the confidence score falls to 0.80, matching white clothing but lagging behind black. At 15 meters, gray clothing exhibits the lowest score of 0.66, reflecting the algorithm's difficulty in distinguishing this color from the background. At 20 meters, gray clothing stabilizes at 0.67, matching black clothing, but by 25 meters, it drops significantly to 0.42.

V. CONCLUSION

The study investigates the performance of objection detection algorithm in identifying pedestrians under critical snowy conditions, which are particularly challenging for automotive safety systems. The focus is on evaluating the ability of the YOLO object detection algorithm to detect pedestrians wearing white, black, and gray clothing across various distances during both daytime and nighttime. The findings highlight key performance trends, showing that the algorithm's confidence in detecting pedestrians decreases as the distance increases, a common limitation in object detection technology. During the day, white clothing consistently offers the highest detection confidence even at long distances, benefiting from its reflective properties, whereas black clothing performs better at night, likely due to specific contrasts in dark or low-light environments. Gray clothing proves to be the most challenging for the algorithm to detect, particularly at longer distances and in low-contrast settings, regardless of the time of day. The study emphasizes the importance of addressing edge cases, like snowy conditions, which can significantly impact detection accuracy and overall safety. These results reveal gaps in current AV testing methodologies, suggesting that while defined sets of test cases can cover many scenarios, they may not adequately account for rare, unpredictable conditions. Therefore, the study encourages for the development of more robust and adaptive testing that can better simulate and handle these real-world edge cases to ensure AVs can perform reliably and safely in all environments.

REFERENCES

- [1] Society of Automotive Engineers standard J3016_202104. https://wiki.unece.org/download/attachments/128418539/SAE%20J3016_202104.pdf?api=v2. Accessed: 07/01/2025.
- [2] Guannan Lou, Yao Deng, Xi Zheng, Mengshi Zhang, and Tianyi Zhang. 2022. "Testing of Autonomous Driving Systems: Where Are We and Where?"
- [3] California Department of Motor Vehicles. Disengagement report. <https://bit.ly/2NmXA1c>. Accessed: 07/01/2025.
- [4] Carmen Gheorghe, Mihai Duguleana, Razvan Gabriel Boboc and Cristian Cezar Postelnicu "Analyzing Real-Time Object Detection with YOLO Algorithm in Automotive Applications: A Review", 31 October 2024. DOI: 10.32604/cmcs.2024.054735
- [5] EUROPEAN NEW CAR ASSESSMENT PROGRAMME (Euro NCAP) TEST PROTOCOL – AEB/LSS VRU systems. <https://www.euroncap.com/media/80156/euro-ncap-aeb-lss-vru-test-protocol-v451.pdf>
- [6] Saeed Rahmani, Sabine Rieder, Erwin de Gelder, Marcel Sonntag, Jorge Lorente Mallada, Sytze Kalisvaart, Vahid Hashemi, Simeon C. Calvert "A Systematic Review of Edge Case Detection in Automated Driving: Methods, Challenges and Future Directions" October 2024.
- [7] Nastaran Moradloo, Iman Mahdinia, Asad J. Khattak, Safety in higher level automated vehicles: Investigating edge cases in crashes of vehicles equipped with automated driving systems, *Accident Analysis & Prevention*, Volume 203, 2024, DOI:10.1016/j.aap.2024.107607.
- [8] K. H. Divya Prasad, M. S. Jincy and S. Ganapathy, "Detection and performance analysis of vulnerable road users in low light conditions using YOLO," 2024 10th International Conference on Communication and Signal Processing (ICCSPP), Melmaruvathur, India, 2024, pp. 862-867, doi: 10.1109/ICCSPP60870.2024.10544095.
- [9] S. Thal et al., "Towards Realistic, Safety-Critical and Complete Test Case Catalogs for Safe Automated Driving in Urban Scenarios," 2023 IEEE Intelligent Vehicles Symposium (IV), Anchorage, AK, USA, 2023, pp. 1-8, doi: 10.1109/IV55152.2023.10186595.
- [10] Shuncheng Tang et al., 2023. A Survey on Automated Driving System Testing: Landscapes and Trends. *ACM Trans. Softw. Eng. Methodol.* 32, 5, Article 124 (September 2023), 62 pages. <https://doi.org/10.1145/3579642>
- [11] X. Zhang et al., "Finding Critical Scenarios for Automated Driving Systems: A Systematic Mapping Study," in *IEEE Transactions on Software Engineering*, vol. 49, no. 3, pp. 991-1026, 1 March 2023, doi: 10.1109/TSE.2022.3170122.
- [12] L. Vater, M. Sonntag, J. Hiller, P. Schaudt and L. Eckstein, "A Systematic Approach Towards the Definition of the Terms Edge Case and Corner Case for Automated Driving," 2023 3rd International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME), Tenerife, Canary Islands, Spain, 2023, pp. 1-6, doi: 10.1109/ICECCME57830.2023.10252672.
- [13] Z. Haimer, K. Mateur, Y. Farhan and A. A. Madi, "YOLO Algorithms Performance Comparison for Object Detection in Adverse Weather Conditions," 2023 3rd International Conference on Innovative Research in Applied Science, Engineering and Technology (IRASET), Mohammedia, Morocco, 2023, pp. 1-7, doi: 10.1109/IRASET57153.2023.10152924.
- [14] D. Bogdoll, M. Nitsche, and J. M. Zöllner, Anomaly detection in autonomous driving: A survey, 2022.
- [15] D. Karunakaran, S. Worrall and E. Nebot, "Efficient statistical validation with edge cases to evaluate Highly Automated Vehicles," 2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC), Rhodes, Greece, 2020, pp. 1-8, doi: 10.1109/ITSC45102.2020.9294590.
- [16] J. Breitenstein, J. -A. Termöhlen, D. Lipinski and T. Fingscheidt, "Systematization of Corner Cases for Visual Perception in Automated Driving," 2020 IEEE Intelligent Vehicles Symposium (IV), Las Vegas, NV, USA, 2020, pp. 1257-1264, doi: 10.1109/IV47402.2020.9304789.
- [17] Y. Xiao, A. Jiang, J. Ye and M. -W. Wang, "Making of Night Vision: Object Detection Under Low-Illumination," in *IEEE Access*, vol. 8, pp. 123075-123086, 2020, doi: 10.1109/ACCESS.2020.3007610.
- [18] J. -A. Bolte, A. Bar, D. Lipinski and T. Fingscheidt, "Towards Corner Case Detection for Autonomous Driving," 2019 IEEE Intelligent Vehicles Symposium (IV), Paris, France, 2019, pp. 438-445, doi: 10.1109/IVS.2019.8813817.
- [19] J. -A. Bolte, A. Bar, D. Lipinski and T. Fingscheidt, "Towards Corner Case Detection for Autonomous Driving," 2019 IEEE Intelligent Vehicles Symposium (IV), Paris, France, 2019, pp. 438-445, doi: 10.1109/IVS.2019.8813817.
- [20] S. Hasirlioglu, I. Doric, C. Lauerer and T. Brandmeier, "Modeling and simulation of rain for the test of automotive sensor systems," 2016 IEEE Intelligent Vehicles Symposium (IV), Gothenburg, Sweden, 2016, pp. 286-291, doi: 10.1109/IVS.2016.7535399. keywords: {Rain;Mathematical model;Safety;Laser beams;Automotive engineering;Laser modes;Sensor systems},
- [21] Bill Canis "Issues in Autonomous Vehicle Testing and Deployment", 2019, Congressional Research Service, Accessed: 07/01/2025